

Lab 1: Conceptual design report

1. Team member roles and responsibilities:

The three members of the team, Adam AIT LAHCEN, Driss BITAR, and Rayane BENYASSINE, worked together on the conceptual design of the database. Every requirement was discussed by the three members, and every modeling decision was approved collectively. Therefore, even though we divided some specific tasks between us, no part of the final model was designed independently.

Driss was mainly responsible for verifying the ER model after our discussions and checking that the different parts of the model remained consistent with the requirements. Adam was mainly responsible for preparing and organizing the PowerPoint presentation of the conceptual design. Rayane was mainly responsible for writing the report and documenting the reasoning behind our choices. However, all of these tasks were rechecked and reconfirmed by the whole group.

2. Key challenges faced and how they were addressed:

One of our main difficulties was deciding when an information should be represented as an attribute, an entity, or a subtype. For the employee hierarchy (PE4, PR3, BD2, MA3), we first considered using a simple status attribute to distinguish Academic and Non-Academic employees. However, after going through the other requirements, we noticed that this distinction was needed in several relationships. For example, Academic is directly involved in supervision, budget management, and project leadership, while Technical staff is involved in maintenance and stock monitoring. This led us to use the multi-level ISA hierarchy :
"Person → Student/Employee → Academic/Non-Academic → Administrative/Technical staff "

We had a similar discussion about the student Program (PE3). We considered first making Program an entity related to Student. However, after examining the rest of the project, we noticed that Program would not have any meaningful relationship with other entities and would not have its own important attributes. We therefore considered it more appropriate to keep it as an attribute of Student.

The reservation approval process (RE1, RE3) was another important difficulty. We initially considered directly connecting Reservation to Academic, but this did not represent clearly what is being approved. The approval concerns the complete reservation context involving the person, the equipment unit, and the research project. We therefore used an aggregation of the Made by relationship and connected this aggregation to Academic through Approved by.

For Maintenance and Calibration (MA1–MA3, CA1–CA4), we considered whether they should be strong entities. We finally modeled them as weak entities because a maintenance record or a calibration record cannot exist independently from the physical Equipment Unit to which it belongs.

Finally, for Consumables (CO4–CO10), we separated procurement from inventory. We used the ternary Contract relationship between Laboratory, Consumable, and Supplier to represent laboratory-specific prices and quantities, while In stock represents what is actually present in a laboratory. We then used aggregation to connect stock information to Technical Staff and to reservations.

3. Justification of the Main Modeling Choices:

For People (PE1, PE2, PE4), we chose ISA specializations instead of representing all categories through attributes. This gives us direct and meaningful endpoints for the different requirements. In particular, Academic is needed for student supervision (PE6), laboratory supervision (LA4), budget management (BD2), and project leadership (PR3), while Technical staff is needed for maintenance (MA3) and inventory monitoring (CO8).

For Laboratories (LA1–LA4), we modeled Laboratory as an entity identified by Lab-id, since laboratories participate in several relationships with equipment units, persons, and academics. Equipment Unit was also kept as an entity because it represents a real and distinguishable physical object that will participate in several later requirements.

For Projects (PR1–PR4), Project is identified by Project Code, and Involves connects persons to projects. We kept Role as an attribute of this relationship because it describes the participation itself, not the person or the project separately.

For Budgets (BD1–BD3), we modeled Budget as an entity and connected it to Academic, Laboratory, and Project through the required relationships. Our previous decision to model Academic as a subtype was useful here because the requirement can directly refer to the appropriate category.

For Equipment (EQ1–EQ7), we distinguished Equipment Model from Equipment Unit, since a model describes specifications while a unit represents a physical item with its own serial number, acquisition date, status, and location. This distinction is also necessary for (EQ5), where a unit must correspond to an equipment model.

For Certifications (CE1–CE6), Requires connects Equipment Model to Certification, while Holds connects Person to Certification. The attributes such as issue date, expiration date, and grade were placed on Holds because they describe a particular certification held by a particular person (CE4).

For Reservations (RE1–RE3), we used a relationship involving Reservation, Equipment Unit, Person, and Research Project. The aggregation used for Approved by allows us to represent that the approval concerns this whole reservation context.

For Maintenance and Calibration, the weak-entity choice reflects their dependence on an Equipment Unit (MA1, CA1). Maintenance is also linked to Technical staff through Carried out by (MA3).

For Consumables (CO4–CO10), we used Contract as a ternary relationship between Laboratory, Consumable, and Supplier. This was important for (CO4), since the price can depend on the laboratory and supplier. The separate In stock relationship then represents the actual inventory of consumables and is used for monitoring and reservation consumption.

4. Requirements Not Expressible in the ER Diagram:

Some requirements could not be fully represented using standard ER constructs. This includes preventing Administrative staff from participating in certain relationships (PR2, RE1), enforcing that PI and Co-PI roles can only be assigned to Academic staff and that each project has exactly one PI (PR3), and checking that the person making a reservation actually holds the certification required by the equipment model (CE7).

The recursive reporting requirement (PE5) also cannot guarantee by itself that the hierarchy is acyclic and ultimately leads to the Dean. In the same way, the arithmetic constraint between money granted and amount disbursed (BD1) and the prevention of overlapping reservations for the same equipment unit (EQ7, RE2) require additional relational-level constraints or database logic.